The Geospatial Equation of Ryan Shtuka.

A Deductive Spatial Analysis and Tactical Recovery Pitch.

Using exclusively environmental, statistical, and spatial data to deduce the highest-probability location of Ryan Shtuka and define the technological toolset required for recovery.



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1:30 AM

Departure from the PLS was originally thought to be 2:10 AM. A crucial text message discovered on a roommate's phone confirms Ryan was preparing to leave the party party at 1:30 AM.

~~**2:10 AM**~~

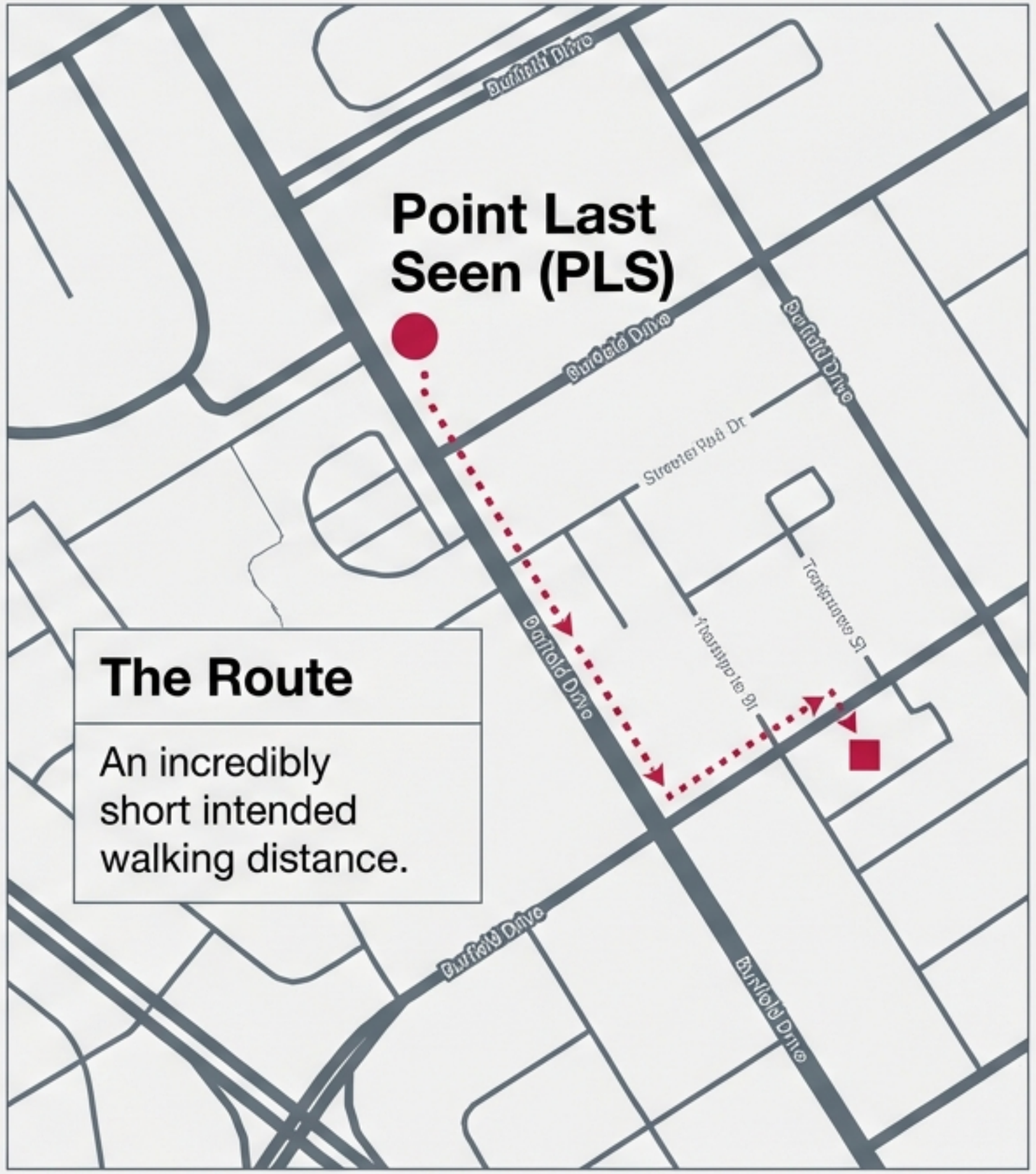
The Subject

Ryan Shtuka.
20 years old, 6’0”, 180 lbs.

Clothing

Blue coat, dark jeans, grey/white shirt, distinctive burgundy ball cap.

3:00 AM



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The Environmental Catalyst



Temperatures

Dropping rapidly from -10°C in the village down to -26°C (-15°F) in the alpine areas.



Precipitation

22 cm (8.6 inches) of fresh snow fell in the 48 hours immediately following his 1:30 AM departure.



Visibility

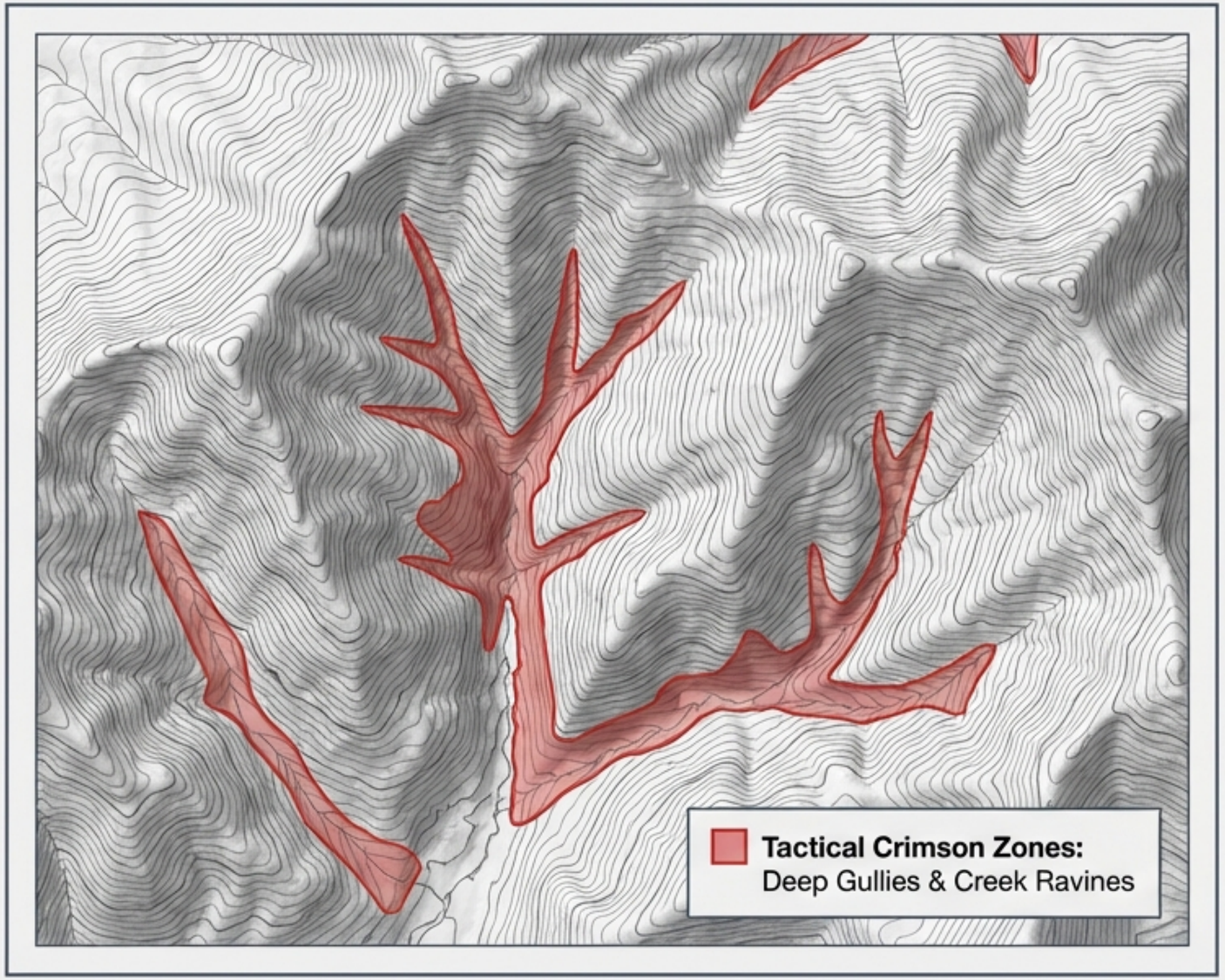
Near-zero visibility on unlit mountain roads.

Forensic Impact

The environment acted as an immediate natural erasure mechanism. The survival window for an exposed individual was measured in mere minutes, drastically limiting the initial radius of foot travel.

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Topography & Natural Hazards



The Terrain Traps

Steep, densely forested creek ravines, tree wells (hidden cavities around tree bases), and deep gullies flank the roads.

SAR Behavioral Statistics

56% to 86% of lost persons in mountainous terrain travel downhill from the Point Last Seen.

SAR Distance Statistics

The median distance a lost person travels ranges from 1 to 2.6 km from the point of origin.

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The Contextual Reality

Law Enforcement Presence

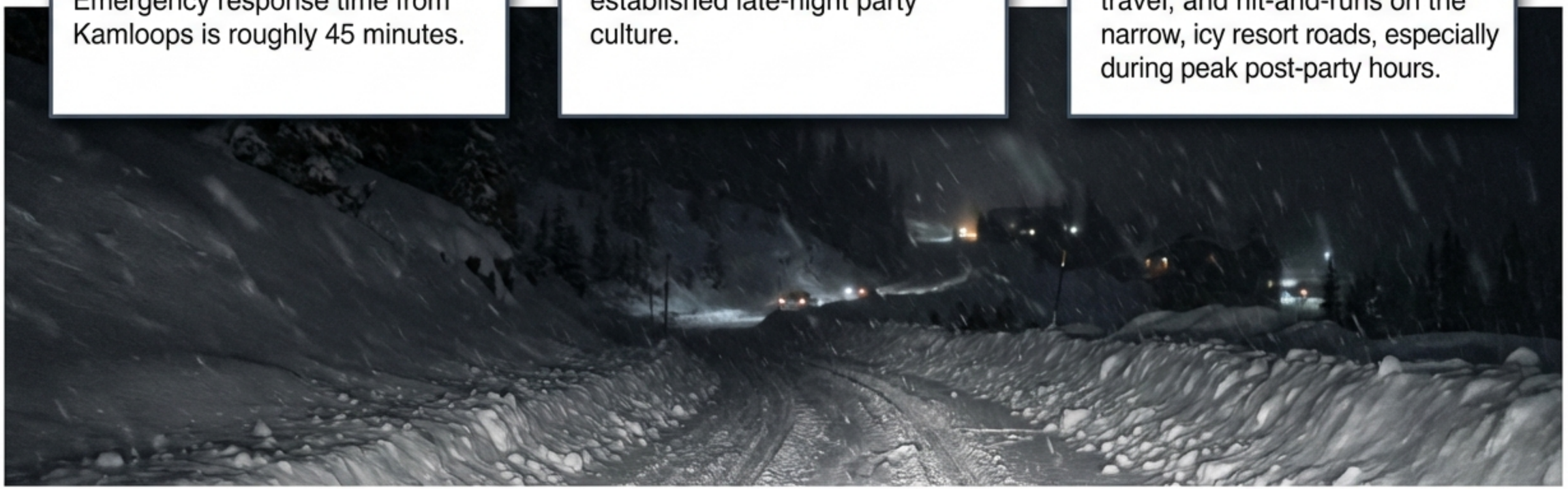
No dedicated local RCMP detachment in Sun Peaks. Emergency response time from Kamloops is roughly 45 minutes.

Local Culture

A heavy transient and seasonal worker population with an established late-night party culture.

Vehicular Risk

Documented local history of reckless driving, high-speed travel, and hit-and-runs on the narrow, icy resort roads, especially during peak post-party hours.



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The Missing Evidence Paradox



The Winter Search

Over 1,000 volunteers, professional SAR teams, drones with infrared heat-signature cameras, helicopters, and 5 highly trained canine units scoured the area.



The Spring Thaw

Record snowpacks melted to reveal bare ground.

The Paradox

If Ryan succumbed to exposure on or immediately adjacent to his walking path, the massive grid searches and the spring melt should have revealed his body, his blue coat, his phone, or his burgundy hat. Absolutely zero trace was found.

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Ruling Out the **Improbable.**

~~Exposure on the Main Path~~ Eliminated.

Unprecedented search efforts and the spring melt mathematically guarantee discovery if remains were on the surface near the route.

~~Voluntary Disappearance~~ Eliminated.

Zero activity on bank accounts, social media, or cell phone since 3:00 AM the night of the disappearance.

~~Deep Wilderness Walk~~ Eliminated.

Deep snowpacks (4 to 6 feet) require specialized snowshoes to navigate. Walking miles into the dense forest in those conditions is physically impossible.

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The Top 1 Possibility

Accidental Vehicular Strike & Concealment

The Alignment

The 1:30 AM departure aligns with peak impaired driving risks. Low visibility and icy roads create the catalyst.

The Concealment

Helvetica Now Text: A panicked driver strikes Ryan, removes the victim from the scene in their vehicle, and drives away. The immediate 22cm snowfall perfectly hides any resulting blood or debris on the road before morning.

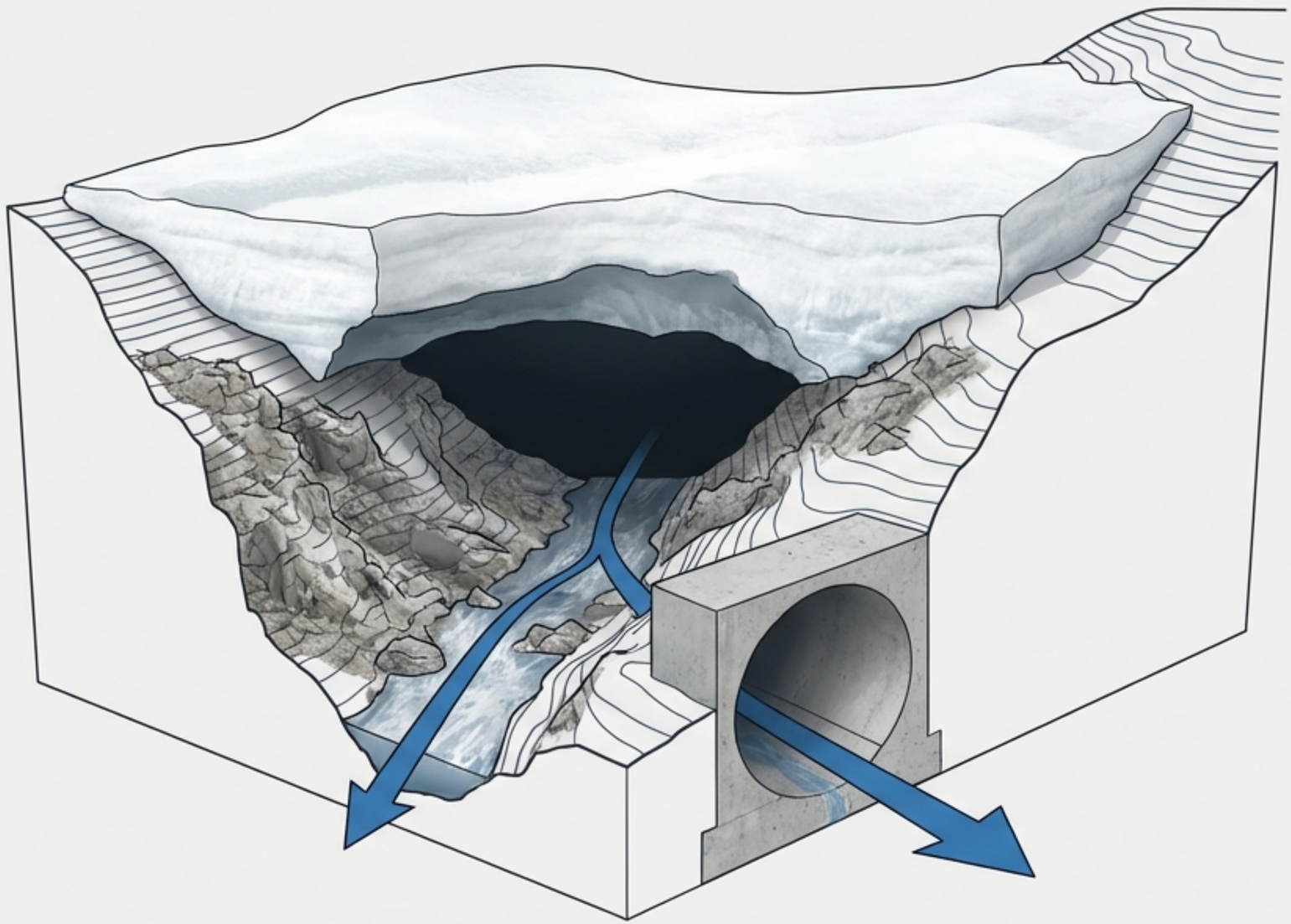
Supporting Data

Helvetica Now Text: An anonymous tip reported shouting (‘get in the car’) near the resort entrance around 2:00 AM, and a single vehicle was spotted leaving the mountain at 4:00 AM.



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Alternative Top Possibility: The Topographical Trap



The Mechanism

If not removed by a vehicle, the only factual alternative is severe downhill disorientation.

The Trap

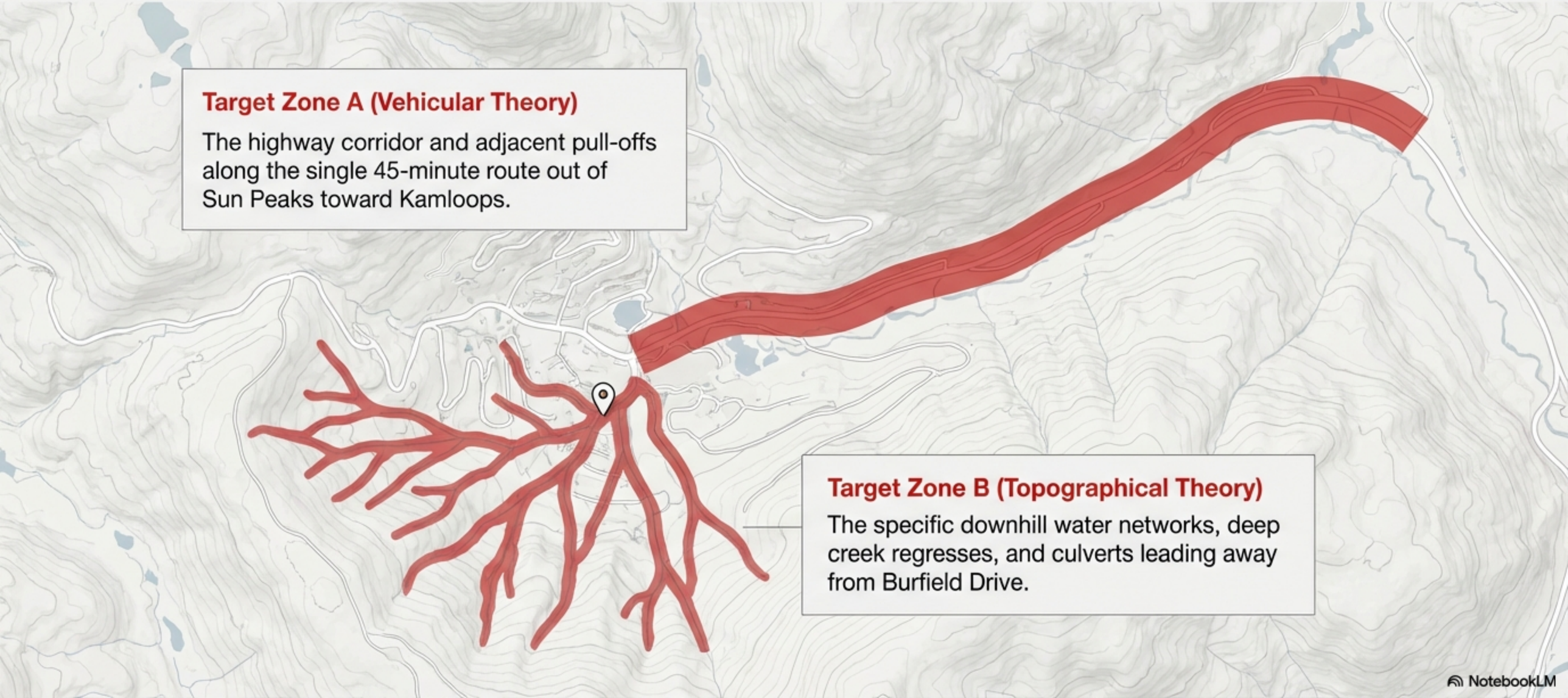
Ryan falls into an inaccessible geographic trap—a deep creek regress, a steep ravine, or a snow-covered culvert.

The Erasure

Spring runoff from the massive snowpack subsequently washed the remains further downhill into a subterranean or water-logged system, preventing discovery during surface searches.

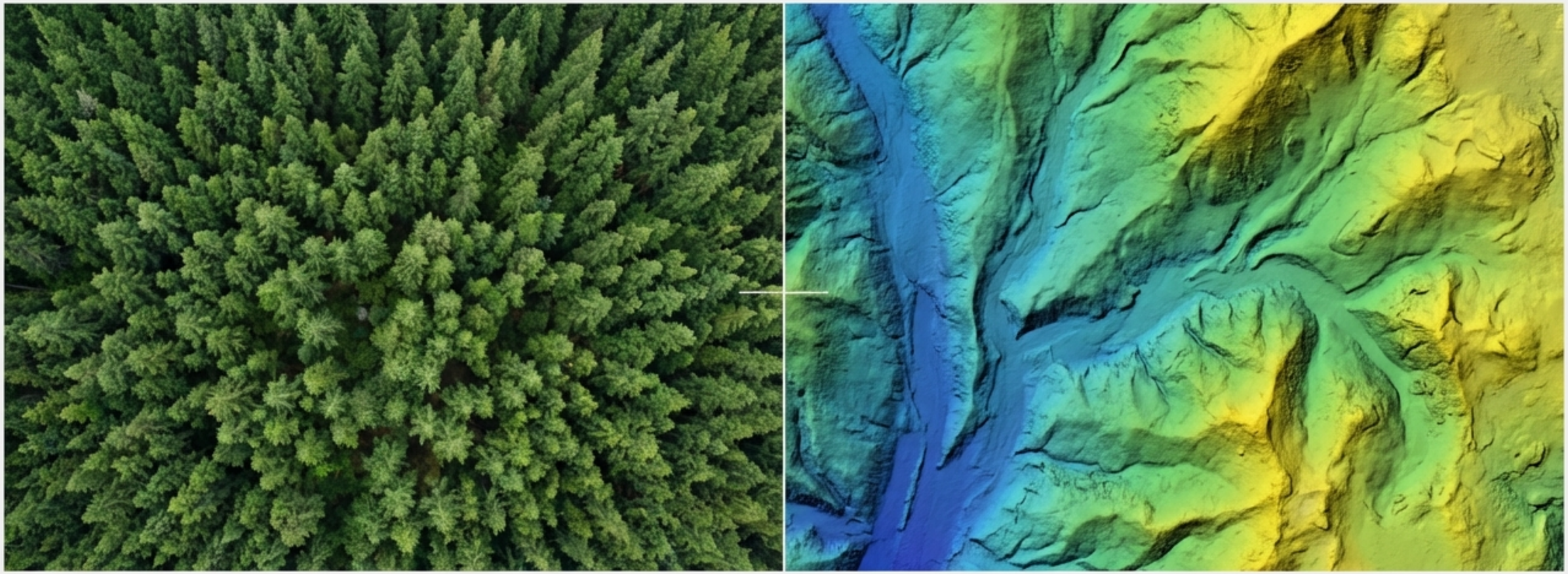
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Spatial Line of Sight & Target Areas



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Airborne & Drone LiDAR



The Capability

Light Detection and Ranging (LiDAR) penetrates dense forest canopy, leaf litter, and snow cover to map the actual ground surface.

The Application

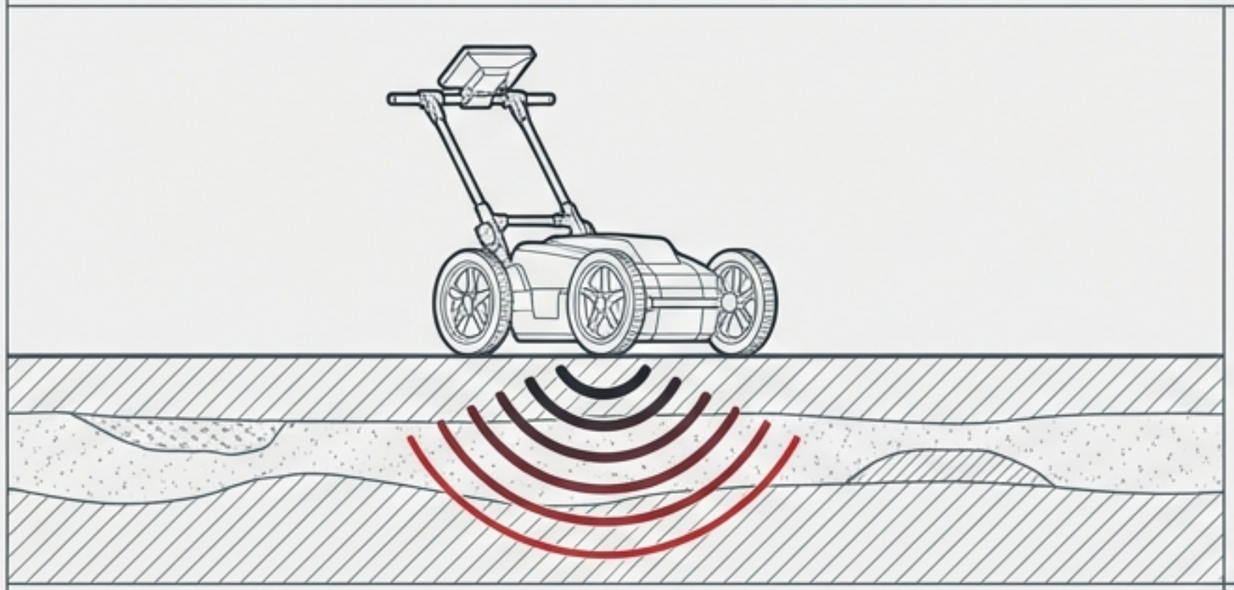
Capable of detecting disturbed soil, unnatural elevation changes (shallow graves), and hidden geological depressions across vast, inhospitable terrain.

Data Source

Leveraging the LidarBC provincial initiative for baseline topographical data.

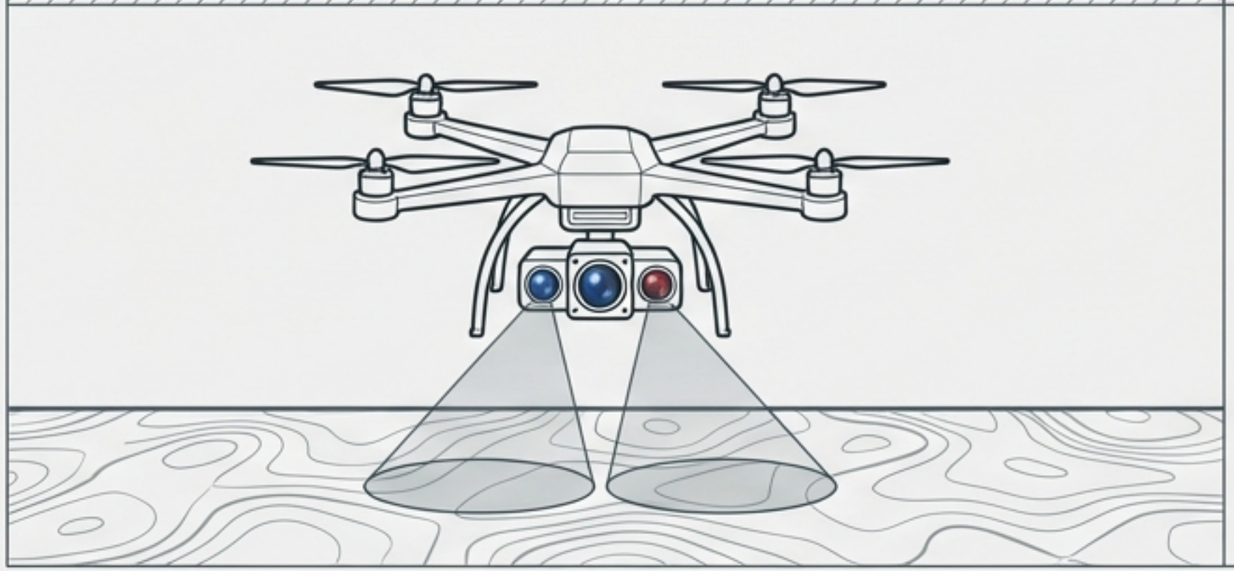
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Ground & Subsurface Technology



Ground Penetrating Radar (GPR)

Deployed to detect anomalies in backfill, buried culverts, and deep sediment where standard visual searches fail.



Multispectral Drones

Utilizing custom software (such as Wings of Mercy color-matching) to systematically scan vast tracts of aerial imagery specifically calibrated for the wavelengths of Ryan’s burgundy hat and blue coat.

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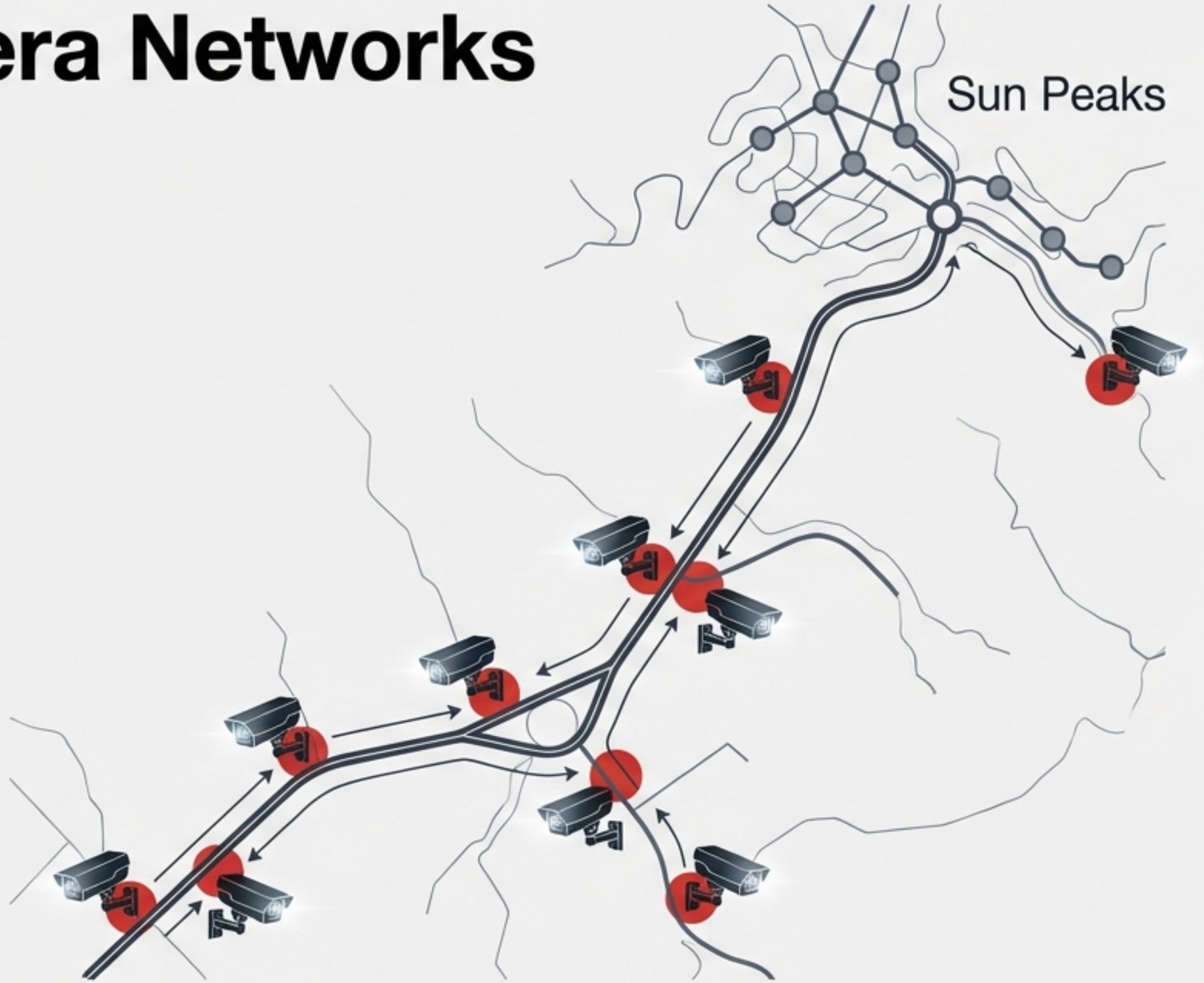
Leveraging Camera Networks

The Chokepoint

Sun Peaks is geographically isolated. There is one primary road out.

The Objective

Retrospective analysis of any surviving traffic cam data, dashcam archives, or dashcam archives, or private property cameras along the route to Kamloops between 1:30 AM and 4:00 AM on February 17, 2018.



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Actionable Next Steps



Step 1: Data Acquisition

Secure BC provincial LiDAR data for the downhill ravines and the highway corridor toward Kamloops.

Step 2: Targeted Deployment

Deploy drone-based LiDAR and GPR exclusively to the specific downhill regresses, culverts, and water outlets identified in Target Zone B.

Step 3: Profile Re-evaluation

Re-evaluate the revised 1:30 AM timeline against historical hit-and-run spatial profiles and the 4:00 AM vehicle sighting.

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The Path to the Truth



Closing Statement

The mountain and the weather erased the immediate physical traces of February 17, 2018. However, by combining objective spatial data with advanced remote sensing technology, we possess the tools to penetrate the wilderness and find the answers.

Contact Info: Kamloops RCMP
Tip Line: 250-828-3000

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